

Storage Unit: My Favorite Problems

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※ Introduction

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※ Algebra

1. (2022 ZeMC) What is the product of the mixed fractions shown below?

$$1\frac{1}{2} \quad 2\frac{2}{3} \quad 3\frac{3}{4}$$

2. (2023 JMPSC) Indigo has six times as many quarters as dimes. The value of her quarters is how many times as large as the value of her dimes?

3. (2024 FLSAM) For how many positive integers T does there exist unique positive integers X and Y for which

$$\frac{2}{71} < \frac{X}{T} < \frac{Y}{T} < \frac{3}{14}.$$

4. (2022 XYZMC) Chad has multiple \$10 dollar bills. He can buy at most 46 socks worth 3 dollars each. At most, how many shoes can he buy if each shoe is worth 12 dollars?

5. (2022 XYZMC) Amy has an equal number of quarters, dimes, nickels, and pennies. If the total cost of all coins is \$8.61, how many coins does she have?

6. (2025 FLSAM) Suppose a , b , and c are real numbers such that

$$ab + bc + ca = 40 - a^2 = 45 - b^2 = 72 - c^2.$$

Find the value of $|abc|$.

7. (2023 JMPSC) Alou has two cylindrical buckets, Bucket One and Bucket Two, which are initially filled with water to depths of 8 cm and 5 cm, respectively. If Bucket Two is fully poured into Bucket One, the depth of Bucket One rises to N cm. If Bucket One is fully poured into Bucket Two, the depth of Bucket Two rises to $N + 3$ cm. Compute N .

8. (2024 FLSAM) Compute the product of all possible values of L such that the polynomial

$$x^4 + 6x^3 + Lx^2 + 12x - 1$$

can be factored into two non-constant polynomials with integer coefficients.

9. (2026 FLSAM) An arithmetic sequence $s_1, s_2, s_3, \dots$ of reals is such that

$$\frac{s_1}{2} = \frac{s_{s_1}}{11} = \frac{s_{s_{s_1}}}{53} \neq 0.$$

Compute s_{10} .

10. (2025 FLSAM) The graphs of $y = x^3 - 5x^2 + 7x + 4$ and $y = 2x^2 - 6$ intersect at A , B , and C . Find the sum of the coordinates of the centroid of triangle ABC .
11. (2026 FLSAM) Quadratic polynomials $P(x)$ and $Q(x)$ with real coefficients are such that the polynomials $P(x)$, $Q(x)$, and $P(x) + Q(x)$ achieve unique minimum values when x is 20, 26, and 25 respectively. Find the x for which $P(x) - Q(x)$ is minimized.

12. (2024 AHMIMS) Real numbers a, b, c , and d satisfy the following system of equations:

$$\begin{cases} a + b + cd = 2 \\ b + c + da = 4 \\ c + d + ab = 5 \\ d + a + bc = 7 \end{cases}$$

Given that $|abcd|$ can be expressed as $\frac{m}{n}$, for relatively prime positive integers m and n , find $m + n$.

13. (2026 FLSAM) Distinct reals a, b , and c satisfy

$$a + b = \sqrt[3]{2a + 2b - 11} + 4$$

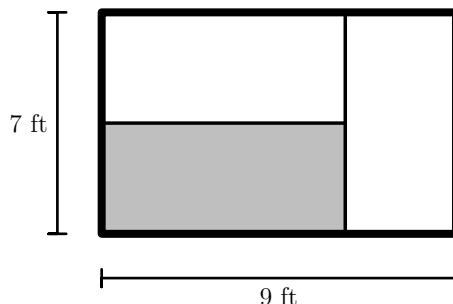
$$b + c = \sqrt[3]{2b + 2c - 11} + 4$$

$$c + a = \sqrt[3]{2c + 2a - 11} + 4$$

Compute $ab + bc + ca$.

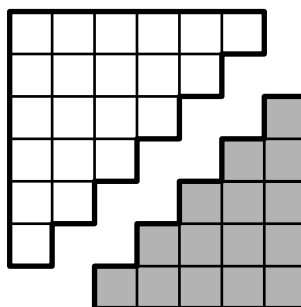
※ Geometry

1. (2024 AHMIMS) Jeffrey's $7\text{ ft} \times 9\text{ ft}$ bedroom window is split into three rectangles of equal area, as shown below.

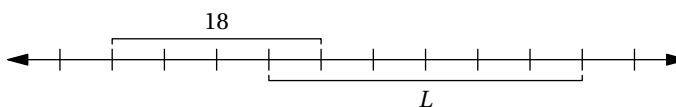


His favorite piece is the shaded rectangle. In feet, what is the perimeter of his favorite piece?

2. (2024 AHMIMS) Hanfei has a square sheet of graph paper with two colors, shaded and unshaded. He then cuts the paper into two parts, as shown below. Given that the area of the shaded region is 40, what is the area of the unshaded region?



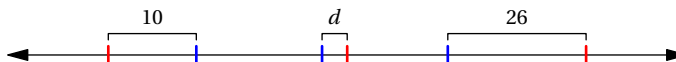
3. (2023 JMPSC) On the following number line, tick marks are equally spaced.



Segments are drawn with lengths 18 and L , find L .

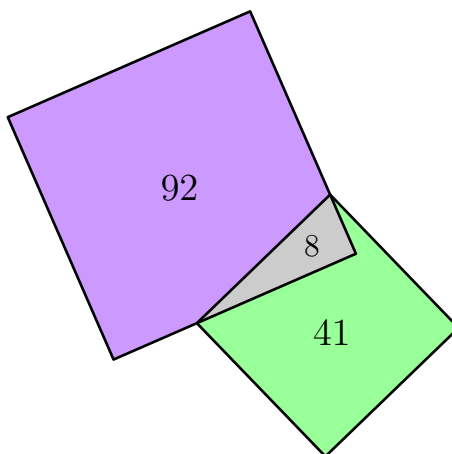
4. (2025 FLSAM) A triangle has side lengths 9, 10, and M . If the area of the triangle is $4M$, find the product of all possible values of M .
5. (2026 FLSAM) Let $ABCDEF$ be a regular hexagon. Suppose P lies on side CD such that $PABC$ and $PDEF$ have areas 20 and 26, respectively. Compute the area of $ABCDEF$.
6. (2026 FLSAM) Let $ABCD$ be a rhombus. Suppose X is a point on segment BD such that $\angle AXC = 90^\circ$. Given that $BX = 20$ and $DX = 26$, compute the area of $ABCD$.
7. (2024 AHMIMS) Intersecting squares $\mathcal{M}$ and $\mathcal{W}$ have side lengths 12 and 14, respectively. The area inside $\mathcal{W}$ and outside $\mathcal{M}$ is twice the area inside $\mathcal{M}$ and outside $\mathcal{W}$. Find the total area covered by both squares.

8. (2025 FLSAM) Rectangle $ABCD$ has side lengths $AB = 24$ and $BC = 21$. Points P and Q lie on BC and DA , respectively, such that $AP = PQ = QC$. Find PQ .
9. (2023 JMPSC) In the unmarked number line below, tick marks of the same color are equally spaced from each other.

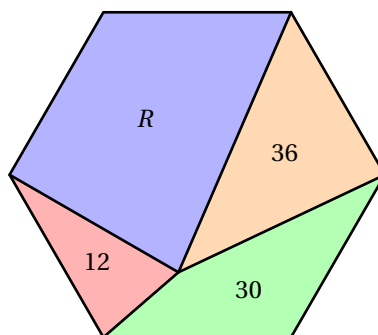


Three distances above are labeled 10, d , and 26. Find d .

10. (2024 AHMIMS) Heptagon $\mathcal{H}$ is formed by two squares and split into three regions with area 8, 41, and 92. Find the perimeter of $\mathcal{H}$.

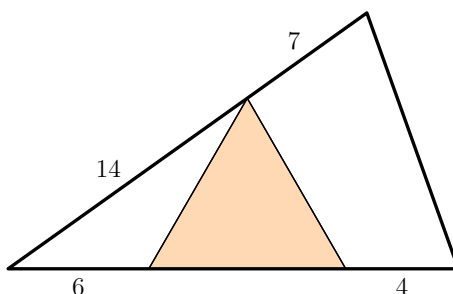


11. (2026 FLSAM) Let $ABCDEF$ be an equiangular hexagon with alternating sides of lengths 8 and 10. A circle, tangent to the sides of length 8, intersects the sides of length 10 at six distinct points. Compute the area of the polygon formed by the six intersection points.
12. (2024 AHMIMS) A regular hexagon is split into polygons with areas R , 12, 30, and 36. Find R .



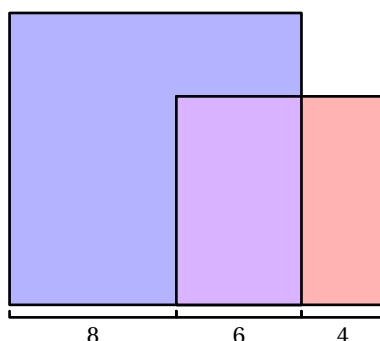
13. (2026 FLSAM) Let ABC be a triangle with orthocenter H such that M and N are midpoints of AB and AC , respectively. Suppose that $MH = 4$, $BH = 6$, and $CH = 8$. Find HN^2 .
14. (2026 FLSAM) Let ABC be a triangle with centroid G . The circle centered at G passing through A intersects segment BC at P and Q such that $BP = 6$, $PQ = 10$, and $QC = 2$. Compute the area of ABC .

15. (2025 FLSAM) In parallelogram $ABCD$, the angle bisectors of $\angle A$ and $\angle B$ intersect segment CD and divide the interior of the parallelogram into four regions: two triangles and two quadrilaterals. Given that the triangles have areas 27 and 75, find the area of $ABCD$.
16. (2023 JMPSC) Let $ABCD$ be a parallelogram with area 140. Points M and N lie on $\overline{AB}$ and $\overline{CD}$ such that $AMCN$ is a rhombus. Given that $MN = 10$ and $AC = 20$, find BD^2 .
17. (2024 FLSAM) In regular polygon $A_1A_2 \cdots A_n$, define X as the reflection of A_1 across A_3 such that X , A_5 , and A_{25} are collinear. Find n .
18. (2024 FLSAM) In rhombus $ABCD$, let M and N be points on $\overline{AB}$ and $\overline{CD}$, respectively, such that parallelogram $AMCN$ has area 240. If $AC = 30$ and $MN = 20$, find the area of $ABCD$.
19. (2026 FLSAM) Let ABC be an acute triangle with orthocenter H and circumcenter O . Suppose lines AO and AH intersect segment BC at X and Y , respectively, such that $BX = 3$, $XY = 5$, and $YC = 2$. Compute the area of $\triangle ABC$.
20. (2025 FLSAM) Let ABC be a triangle with centroid G . Suppose the reflection of A across B lies on the circumcircle of BGC . Given that $GB = 3$ and $GC = 9$, find the area of ABC .
21. (2025 FLSAM) Let ABC be a triangle with points P and Q on segment BC with B, P, Q, C occurring in that order such that
- $$\angle BAP = \angle PAQ = \angle QAC = 45^\circ.$$
- Given that $BP = 40$ and $QC = 60$, compute the area of ABC .
22. (2024 AHMIMS) An equilateral triangle is placed inside a larger triangle splitting one side into lengths of 14 and 7 and another side into lengths of 6 and 4, as shown below.

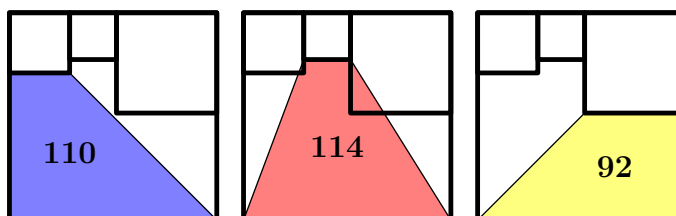


If the area of the larger triangle can be written as $m\sqrt{n}$, find $m + n$.

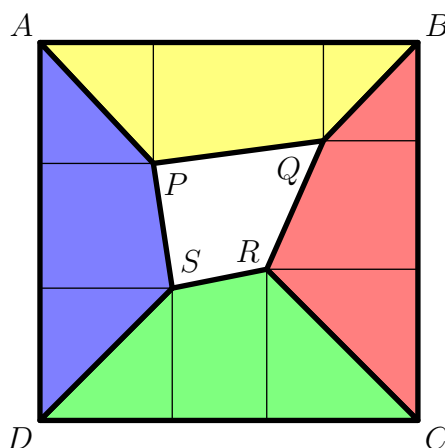
23. (2023 JMPSC) Two overlapping squares form a hexagon, with an area A , such that segments have lengths 8, 6, and 4. Compute A .



24. (2022 XYZMC) An equilateral triangle with side length 12 is placed in space. Two points, X and Y are so placed such that the distances from both the points to each of the vertices are 12, 12, and 6 respectively. Find the sum of all distinct values of XY^2 .
25. (2026 FLSAM) Let ABC be an acute triangle with circumcenter O and altitudes BP and CQ such that $PQ = 12$ and $BC = 20$. Suppose that AO intersects PQ at R with $\frac{AR}{AO} = \frac{3}{4}$. Compute the area of ABC .
26. (2022 XYZMC) Let $ABCD$ be a trapezoid with $\overline{AB} \parallel \overline{DC}$ and $\angle C$ and $\angle D$ acute. Suppose $ABCD$ has an inscribed circle with center O and radius 2. If $AD = 5$ and the area of $\triangle BOD$ is 2.5, find the length of BC .
27. (2025 FLSAM) Acute triangle ABC has circumcenter O and circumradius 81. Let line ℓ be the tangent to the circumcircle of ABC at A . Given that the distances from B and C to ℓ are 64 and 144 respectively, find the area of ABC .
28. (2025 FLSAM) Convex quadrilateral $ABCD$ has diagonals $AC = 72$ and $BD = 78$ which intersect at I . Let P , Q , and R be the circumcenters of triangles BIC , CID , and DIA respectively. It is given that $QI \perp AB$, $PI = 8$, and $RI = 32$. Compute the square of the area of $ABCD$.
29. (2022 XYZIME) The line $y = -2x$ is rotated clockwise about the point $(14, 2)$ to $y = mx$, where $m \neq -2$. If m can be expressed as $\frac{a}{b}$, where a and b are relatively prime positive integers, find $a + b$.
30. (2022 XYZIME) A square contains 3 smaller squares to form three trapezoids of area 110, 114, and 92 as shown. If the area of the largest square can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers, find $m + n$.



31. (2023 ZeMC) Four squares are placed in each corner of a larger square, as shown in the diagram below. If the areas of quadrilaterals $APQB$, $BQRC$, $CRSD$ and $DSPA$ are 20, 23, 26 and 25, respectively, what is the area of quadrilateral $PQRS$?



32. (2026 FLSAM) Let ABC be a triangle with $AB = 24$. Points P and Q lie on segments CA and CB such that $PA = AB = BQ$. Suppose AQ and BP intersect at R . Points S and T lie on the circumcircles of ARP and BRQ such that $AS \parallel CB$ and $BT \parallel CA$. Given that $SR = 16$ and $RT = 20$, compute the area of ABC .

※ Number Theory

- (2025 FLSAM) Jason's favorite two-digit number is such that subtracting 2 or 4 from it results in a number that shares exactly one digit with the original, but subtracting 3 results in a number that shares no digits with the original. Find Jason's favorite number.
- (2022 JMPSC) There exist three digits a , b , and c for which the two-digit positive integers $\overline{ab}$, $\overline{bc}$, and $\overline{ca}$ sum to a perfect square. What is $a + b + c$?
- For a positive integer n , let $s_{10}(n)$ be the sum of the digits of n . Compute the product of all distinct possible values of $s_{10}(n + 6) - s_{10}(n)$ over all two-digit positive integers n .
- (2022 XYZIME) A positive integer N has exactly 247 composite divisors. Find the sum of the possible number of divisors of N .
- (2026 FLSAM) Find the unique positive integer N such that
 - exactly 12 divisors of N are divisible by 12, and
 - exactly 15 divisors of N are divisible by 15.
- (2024 FLSAM) Sasha's favorite number satisfied the following conditions:
 - Exactly 4 distinct divisors of her favorite number have 4 factors.
 - Exactly 6 distinct divisors of her favorite number have 6 factors.

Find the minimum possible value of Sasha's favorite number.

- (2025 FLSAM) Primes p and q are such that $\frac{3q+4}{p}$ and $\frac{4p+3}{q}$ are integers. Compute the sum of all distinct possible values of pq .
- (2025 FLSAM) Find the smallest positive integer n such that

$$3 \left\lfloor \frac{n}{3} \right\rfloor - 4 \left\lfloor \frac{n}{4} \right\rfloor = 7 \left\lfloor \frac{n}{7} \right\rfloor - 8 \left\lfloor \frac{n}{8} \right\rfloor = 11 \left\lfloor \frac{n}{11} \right\rfloor - 12 \left\lfloor \frac{n}{12} \right\rfloor = 1.$$

※ Combinatorics

1. (2025 FLSAM) Ten people are lined up in a row, three of which are Aaryan, Ashvin, and Darsh. Given that Aaryan stands next to Ashvin, what is the probability Ashvin also stands next to Darsh?
2. (2026 FLSAM) Twelve distinguishable socks, four of each of the colors red, blue, and green, are randomly placed into three bins with each bin receiving four socks. Compute the probability that each bin has exactly one pair of identically colored socks.
3. (2025 FLSAM) Find the number of six-digit positive integers N which satisfy the following properties:
 - N is divisible by 2 and exactly two of its digits are 2; and
 - N is divisible by 3 and exactly three of its digits are 3.
4. (2024 FLSAM) Each of Jason's socks have either three holes or five holes. If Jason randomly selects a hole from any of his socks, the probability that this hole comes from a sock with three holes is $\frac{3}{14}$. If the probability of Jason selecting a sock with three holes is $\frac{p}{q}$ for relatively prime positive integers p and q , find $p + q$.
5. (2026 FLSAM) Let $\mathcal{A} = (1, 1, 1, 1, 2, 2, 2, 3, 3, 4)$. For a permutation of $\mathcal{A}$, let its *quintessentiality* be the number of entries of the permutation (before the final entry) that are at most the next entry. Let m be the minimum possible quintessentiality of a permutation of $\mathcal{A}$. Find the number of permutations of $\mathcal{A}$ with quintessentiality m .
6. (2025 FLSAM) Find the number of 5-tuples (F, L, S, A, M) of digits such that $F \cdot L, L \cdot S, S \cdot A$, and $A \cdot M$ all have units digit 6.
7. (2025 FLSAM) The positive divisors of 1001 are randomly written on four two-sided cards, one on each face. Find the probability that on every card, one number divides the other.
8. (2026 FLSAM) Find the number of ways to place 9 standard six-sided dice onto a 3×3 grid such that for every pair of orthogonally adjacent dice, their touching faces have the same number.
9. (2026 FLSAM) Ten apples of distinct sizes are randomly arranged in a circle. An alligator is first located at the smallest apple and eats it. Every second from there, the alligator looks at the two apples adjacent to him and eats the larger one, repeating until there is only one apple left, which he also eats. Compute the probability that the alligator eats the second largest apple before the largest apple.
10. (2026 FLSAM) Jason places each of his 100 marbles numbered 1 to 100 into one of two distinguishable jars so that each jar has at least one marble. Compute the number of ways he can do this so that the product of the numbers in each of the two jars are relatively prime to each other.
11. (2026 FLSAM) Ten identical pandas sit in a row of 14 chairs such that each chair is occupied by at most one panda. Each panda rolls either two chairs to the left or two chairs to the right. Compute the number of initial positions of the pandas such that it is possible for the pandas to occupy distinct chairs at the end.
12. (2025 FLSAM) Find the number of sequences $s_1, s_2, \dots, s_{15}$ such that $s_1 = 1, s_2 = 2, s_3 = 3$, and for all integers $3 < k \leq 15$,

$$s_k \in \{s_{k-1}, s_{k-2}, s_{k-3}\}.$$